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Research on Optimization of Charging Strategy Control for Aged Batteries

Jinlei Sun , Member, IEEE, Qian Ma, Chuanyu Tang, Tianru Wang, Tao Jiang, and Yong Tang

Abstract—Rapid charging of lithium-ion batteries has become a major bottleneck for the large-scale deployment of electric vehicles. In this article, a charging optimization method is proposed for lithium ion batteries based on their aging characteristic, which takes charging time and temperature rise into consideration. An electric-thermal battery model is utilized to estimate the battery temperature during charging. A search for the optimal charging current pattern is performed using a particle swarm optimization (PSO) algorithm combined with fuzzy reasoning. The experiments are conducted on batteries with different aging states to verify the proposed method. Besides, the proposed method is compared with traditional constant current-constant voltage (CC-CV) charging method. Results show that the proposed charging strategy has advantages on maximum temperature rise maintenance and charging speed improvement for aged batteries with unknown aging state. The proposed charging method provides a way for aged batteries charging considering charging time and temperature rise.

Index Terms—Aging, charging optimization control strategy, lithium-ion battery, particle swarm optimization.

I. INTRODUCTION

THE development of industrial society has accelerated the deterioration of the environment while consuming nonrenewable energy. Finding clean and efficient renewable energy is an important means to alleviate the energy crisis and environmental problems. Lithium-ion batteries are widely used in the fields of electric vehicles, power storage systems and mobile electronic devices due to their high energy density, low self-discharge rate, no memory effect [1], [2].

Charging is the most common mode of battery operation. To pursue a better charging experience, it is often expected that increasing the charging power will shorten the charging time. However, the increase in charging power will accelerate degeneration in battery capacity and lead to a decline in the battery cycle life [3]. At the same time, an increase in the charging current will also lead to an increase in battery heat generation. Especially under serious conditions, it will cause thermal runaway and other safety problems [4].

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Numerous charging methods have been reported in the literature in order to reduce charging time and maximize battery life. Thanh *et al.* and Li Jiang *et al.* used the Taguchi method to optimize the charging current of the multistage constant-current charging method, which has the potential to improve charging speed and energy efficiency [5]. Zhang used a dynamic programming (DP) optimization algorithm to achieve the balance between charging loss and time [6]. Eduardo Perez *et al.* proposed an optimal charging method based on coupled electro-thermal-aging dynamics, and multiobjective optimal control is utilized to achieve the optimally tradeoff between charge time and health degradation [7]. Liu *et al.* extended battery life using the single particle model (SPM), and the effect of side reactions on lithium evolution in the charging process was taken into consideration [8]. Zhang *et al.* realized a balance between charging time and charging temperature rise by using the genetic algorithm combined with the analysis of battery polarization characteristics [9]. Ahmet Aktas designed a charging circuit and proposed a charging method considering temperature rise for military vehicle [10].

The existed charging optimization methods can be categorized as either charging current improvement based methods or battery model based methods. The former methods utilized a variety of artificial intelligence algorithms to optimize the charging performance, such as the genetic algorithm, dynamic programming algorithm and pseudospectrum algorithm. However, they often lack of theoretical foundations in choosing the optimal charging currents. Battery model based methods can employ electrochemical model or equivalent circuit model to search for optimal charging current. Nevertheless, most of these methods focus on the optimization of charging strategies with a fresh battery or one with a specific aging state. There is little research on charging optimization of aged battery. For an aged battery, the internal resistance increases gradually with aging, and the temperature rise during charging should be considered during charging. If constant current value is utilized to charge both aged and fresh batteries without considering the temperature changes caused by internal resistance variation and capacity degeneration, the battery may work at high temperature and suffer from accelerated aging. It has been indicated that these temperature changes directly affect battery aging and performance [11]. Therefore, it is necessary to adjust the charging strategy according to different aging states during battery usage considering both temperature rise and charging speed.

This article focuses on charging optimization strategy for the lithium-ion battery with different aging states. There are

TABLE I
NOMINAL SPECIFICATIONS OF THE ISR18650-2.6Ah BATTERY

Item	Value
Capacity	2.6 Ah
Charging cutoff voltage	4.2 V
Discharging cutoff voltage	2.75 V
Nominal voltage	3.6 V
Working temperature	0 °C–45 °C

three main contributions. First, a battery electric-thermal model is established to estimate battery temperature variation during charging. Second, a charging strategy considering temperature rise and charging speed is proposed for battery with different SOHs. Third, the charging current pattern is optimized using particle swarm optimization (PSO) algorithm combined with fuzzy reasoning to search for the optimal results.

The remainder of the letter is organized as follows. Section II analyzes the battery parameter variations for aged batteries through experiments. Section III proposes a charging strategy for aged batteries. Section IV shows the experimental results and the comparisons with the constant current-constant voltage (CC-CV) method. Finally, Section V concludes this article.

II. BATTERY AGING EXPERIMENTS

A. Battery and Devices

To investigate the aging mechanisms of a lithium-ion battery and to obtain battery parameters at various states of health (SOHs), cycle tests were conducted under constant current 1C. The parameters of ISR18650-2.6Ah (NMC, 18650) are listed in Table I.

A multichannel battery testing system NEWARE-BTS4000 (5 V, 200 A, four channels, Neware Co. Ltd., Shenzhen, China) was used to record the voltage and current during charging and discharging. The battery testing system could measure voltage with an accuracy of $\pm 0.1\%$ and has four measurement ranges, 0.05A–10A, 10A–50A, 50A–100A, and 100A–200A with an accuracy of $\pm 0.1\%$ for each measurement range. A GDW-50 (Taisite Experimental Equipment Co. Ltd., Nanjing, China) thermal chamber is used to maintain the ambient temperature. The temperature range of the thermal chamber is $-50\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$. In addition, a TC-08 temperature recorder is used to monitor the surface temperature of the battery. The temperature range is $-270\text{ }^{\circ}\text{C}$ to $400\text{ }^{\circ}\text{C}$ with an accuracy of $\pm 0.1\text{ }^{\circ}\text{C}$.

B. Testing Method

To provide data support for the electric-thermal model of the battery, three screened batteries are selected to conduct tests, and the tests can be divided into cycle aging tests and characteristic tests, the former is performed at a 1C constant current rate, and the latter includes a capacity test, hybrid pulse power characteristic (HPPC) test, and incremental capacity analysis (ICA) test. The experimental flowchart is shown in Fig. 1. The tests are conducted in a thermal chamber to maintain the ambient temperature at $25\text{ }^{\circ}\text{C}$. The constant current cycle test is widely used for battery aging research [12].

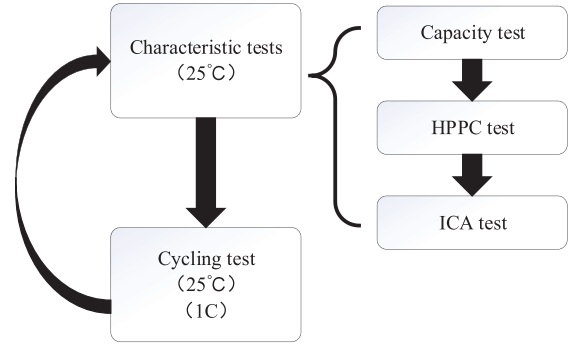


Fig. 1. Experimental flowchart.

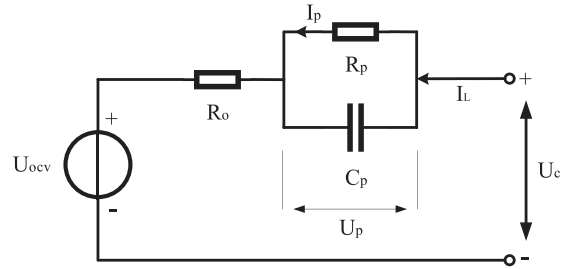


Fig. 2. Thevenin model.

The HPPC test is conducted to obtain the battery characteristic parameters. A 10 second charging pulse with a 1C constant current rate is applied first, and then the battery rests for 40 seconds. Finally, a 10 second discharging pulse with a 0.75C constant current is applied.

Then, the battery is charged to reach another state of charge (SOC) status. The same procedure continues until the charging cutoff voltage limit is reached. A set of characteristic tests is conducted after every 50 cycles.

C. Battery Parameters at Various SOHs

During the aging process, the capacity of a lithium-ion battery gradually degenerates from beginning of life (BOL) to end of life (EOL). Therefore, the capacity at the time of BOL or the actual maximum rechargeable capacity under the current aging state should be considered when calculating SOC. The SOC mentioned in this article is defined as the ratio of the current charge to the fully charged capacity. The equivalent circuit model approach uses the circuit elements to represent the dynamic performance of the battery to avoid complex calculations. In this article, the Thevenin model is chosen to balance calculation accuracy and computational effort [8]. The structure of the Thevenin model is shown in Fig. 2.

Where U_{ocv} is the battery open circuit voltage (OCV); U_c is the battery terminal voltage; I_L is the battery charge current (it is assumed to be positive for charging and negative for discharging); R_o is the ohmic resistance; R_p is the polarization resistance; C_p is the polarization capacity; and I_p and U_p are the polarization current and polarization voltage on the R_p , respectively. The parameters of the Thevenin equivalent circuit model are identified using the least squares method (LSM).

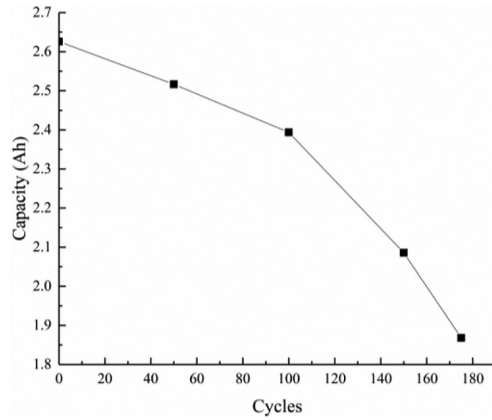


Fig. 3. Capacity degeneration curve at different cycles.

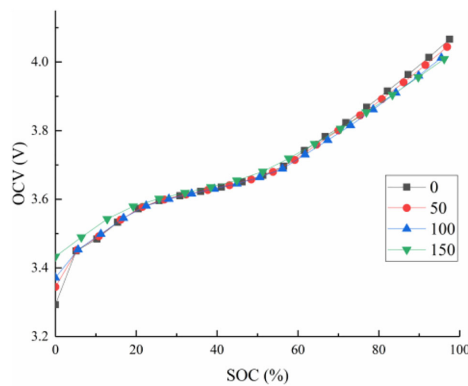


Fig. 4. OCV curves at different cycles.

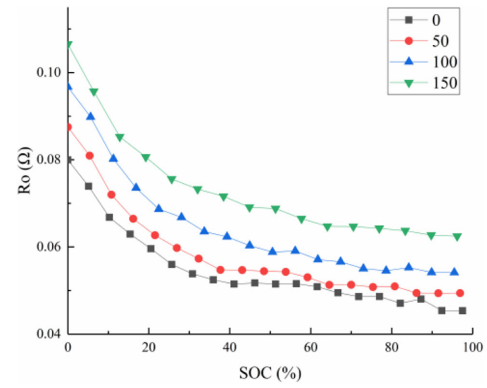


Fig. 5. Ohmic resistance at different cycles.

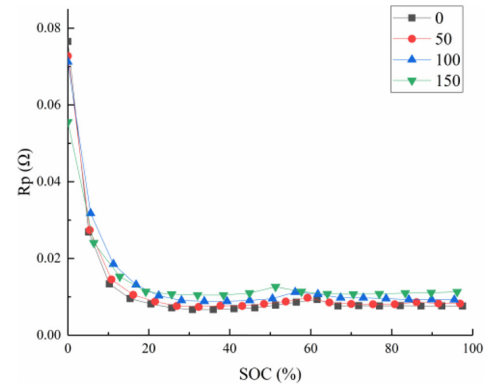


Fig. 6. Variation in polarization resistance with aging.

The variation trend of characteristic parameters such as OCV curves, ohmic resistance and polarization resistance are selected for analysis. The battery parameters at various SOHs identified through HPPC tests are as follows.

1) *Capacity*: As is shown in Fig. 3, the capacity tends to decrease with the increase of cycling. After the first 100 cycles, the battery capacity degenerates from 2.63 Ah to 2.4 Ah. By the 175th cycle, the battery capacity decreases to 1.84 Ah. The battery capacity degenerates continuously with the increase cycle number, and the degeneration speed increases gradually after 100 cycles.

2) *OCV Curves*: OCV is a function of SOC, which is usually used for SOC estimation. The curve is shown in Fig. 4, the shape of the OCV curve is almost the same, regardless of the number of cycles. The curves for OCV within the whole SOC range almost coincide, indicating that the OCV curve is independent of aging cycles.

3) *Ohmic Resistance*: As is shown in Fig. 5, the difference of ohmic resistance is obvious for each measured cycle point within the SOC range from 0% to 100%. The ohmic resistance decreases as the SOC range increases. Meanwhile, the overall ohmic resistance tends to increase with the increase of cycle number. It increases dramatically after 100 cycles.

4) *Polarization Resistance*: Fig. 6 shows the changing trend of polarization resistance for different cycles within the SOC range from 0% to 100%. The polarization is extremely obvious

at the 0% SOC point. That explains why the battery terminal voltage increases rapidly when charging commences. The polarization resistance is smooth from 20% SOC to 100% SOC.

It can be seen that battery capacity and ohmic resistance change significantly with the various aging states from the analysis above. Variations in OCV and polarization resistance caused by aging are not obvious. Using the same charging current for both aged and fresh batteries cannot fully consider the temperature changes caused by the variation of internal resistance and capacity for aged batteries. It has been proven that these temperature variations directly affect aging and performance. Therefore, in order to develop charging strategies for different aging stages, the variations of battery parameters during aging needs to be understood.

D. State Estimation of Aged Battery

In this section, ICA is used to investigate the battery aging mechanisms first, and then a principal component analysis (PCA) based state estimation method is proposed to determine capacity and resistance of aged battery using the characteristic variations in ICA curves.

Incremental capacity (IC) curves are calculated by using the derivative of the voltage profile, and this profile can reflect the evolution of electrochemical processes during charging [13]. Therefore, IC curves are usually used for the identification of battery capacity degradation mechanisms and state of health

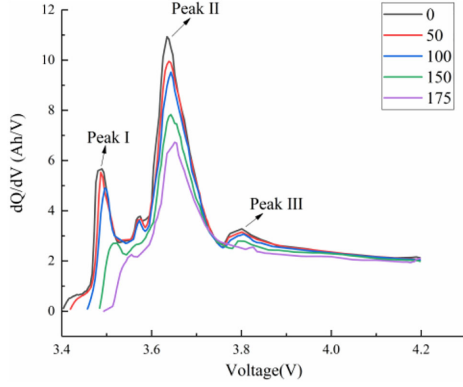


Fig. 7. The evolution of the IC curve along with battery cycling at various SOHs.

monitoring. Peaks in the IC curve can reveal phase transition phenomena in the battery, and [14] has proved that the peak area can be used to estimate the capacity degradation. In this article, the evolution of the value and position of the peaks are used to identify battery aging mechanisms. Fig. 7 shows the evolution of the IC curve along with battery aging, which is derivative from the 1/20C charging process during ICA test. There are three distinct peaks (Peak I, II, III) for each aging state in the curve. The amplitude and position of each peak change significantly with battery aging, which reflects the loss of active electrode materials and lithium inventory [15]. Therefore, the amplitudes and positions of three peaks for each aging state can be used to estimate the state of battery. And the estimation results provide support for charging strategy of aged battery.

In order to quantitatively estimate the capacity and internal resistance of battery using ICA curve, the PCA method is utilized. The standard expression is shown as (1).

$$y = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 \quad (1)$$

where y is the resistance or capacity at various SOHs; x_1 and x_2 are the value and position of Peak I; x_3 and x_4 are the value and position of Peak II, respectively; x_5 and x_6 are the value and position of Peak III, respectively; and a_1 to a_6 are the parameters to be determined. Capacity and resistance are determined by linear regression under different aging conditions, and the results are as follows:

$$\begin{aligned} Q &= -0.32x_2 - 227.52x_3 + 147.14x_5 + 0.8152x_6 + 269.96 \\ R &= 0.2854x_1 - 0.004x_2 - 0.0282x_6 - 0.8276 \end{aligned} \quad (2)$$

where Q is the maximum available capacity of the battery charged at 0.05C; R is the lumped resistance of the battery, which is equal to the average value of the sum of ohmic internal resistance and polarization internal resistance in the full SOC range. Fig. 8 shows the results of estimation.

As is shown in Fig. 8, the error of the capacity estimation is within 0.2 Ah and the error of lumped resistance is within 10 mΩ. The variation trend of the model estimation results is the same as the experimental results.

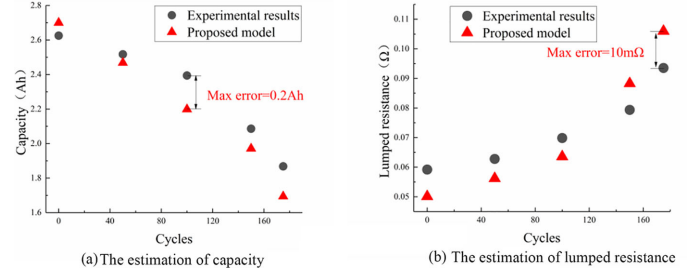


Fig. 8. The parameter estimation results.

III. CHARGING STRATEGY FOR AGED BATTERIES

A. Electric-Thermal Battery Model

To optimize the current in the charging strategy, an accurate and reliable battery model is necessary. The battery voltage and temperature can be calculated according to this model. Using the Thevenin model mentioned in Section II, these can be expressed as eq. (3).

$$\begin{cases} \frac{dU_p}{dt} = -\frac{1}{R_p C_p} U_p + \frac{1}{C_p} I_L \\ U_c = U_{ocv} + I_L R_o + U_p \\ \tau_c = R_p C_p \end{cases} \quad (3)$$

where τ_c is the time constant of the RC network. eq. (3) can be obtained by using discretization:

$$\begin{aligned} U_c(n) &= \\ U_{ocv}(n) + I_L(n)R_o + \frac{TR_p}{\tau_c} I_L(n-1) + \left(1 - \frac{T}{\tau_c}\right) U_p(n-1) \end{aligned} \quad (4)$$

where $U_c(n)$, $U_{ocv}(n)$, $I_L(n)$ and $U_p(n)$ are values of U_c , U_{ocv} , I_L and U_p at the n -th second, respectively. T is the sampling time.

To estimate the battery temperature variation during charging, it is necessary to model the battery thermal dynamics. The thermal model of the battery consists of two parts, thermal generation and thermal release. In this article, the internal temperature distribution of the battery is assumed to be uniform.

In our previous study [16], a battery temperature calculation method based on the Bernardi heating model [17] was proposed, which can be written as:

$$\begin{aligned} T_c(n) &= \\ T_c(n-1) + \frac{I_L^2 R_o + I_L U_p + I_L T_c \frac{dU_{ocv}}{dT_c} - hS(T_c - T_a)}{Cm} \end{aligned} \quad (5)$$

where T_c is the average battery surface temperature; h is the heat transfer coefficient; S is the total superficial area of battery; T_a is the ambient temperature; C represents the heat capacity of the battery; m is the battery mass; and $T_c(n)$ is the value of T_c at the n -th second.

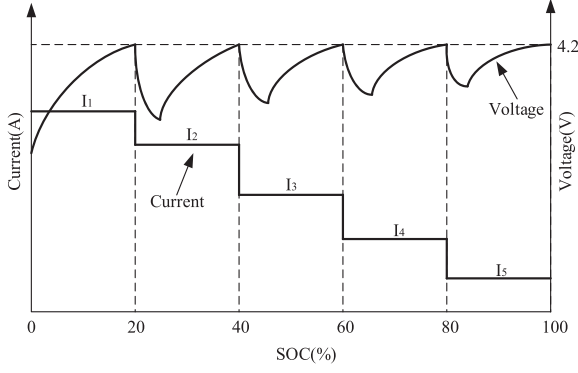


Fig. 9. The schematic of the five-stage constant current charging method.

B. Problem Formulation

In this article, the five-stage constant-current charging approach [18] is selected to perform lithium-ion battery rapid charging. When SOC is low, current rate is large to enhance the charging speed, and the current rates decreases gradually to reduce the effect of overpotential caused by internal resistance. Ref. [19] explores the influence of the charging effect by changing the number of current segments in the charging process. The experimental results show that when the current segment exceeds five stages, the charging effect hardly improves, but it increases the computational load and slows down the convergence speed of the algorithm. The schematic of five-stage constant current charging is shown in Fig. 9.

In Fig. 9, the whole charging process is divided into 5 segments, and the current of each segment gradually decreases. The current rate at first is set large to reduce charging time, and then the current is decreased to reduce the chemical reaction stress and temperature rise during charging [20]. The charging current is set correspondingly in each charging stage during charging, which indicates 20% SOC. The SOC value of the final stage is recorded as SOC_{max} , and the charging SOC range is from SOC 0% to SOC_{max} .

The charging time for each step can be given as:

$$t_k = \frac{Q_k}{I_k} = \frac{\Delta SOC_k \cdot Q}{I_k}, k = 1, 2, \dots, 5 \quad (6)$$

where t_k is the charging time for the k-th step; Q_k is the charged capacity in the k-th step; I_k represents the charging current for the k-th step; and Q is the maximum capacity obtained by estimation under the current aging state.

In the final stage, the charging process usually fails to reach the maximum capacity Q due to battery polarization. Therefore, the charging time of the fifth stage should be calculated separately. The total time of the whole charging process is calculated by:

$$t = \sum_{k=1}^4 t_k + t_5 = \sum_{k=1}^4 t_k + \frac{Q_5}{I_5} \quad (7)$$

where t is the total time of the whole charging process; t_5 and I_5 are the charging time and current of the fifth stage, respectively; and Q_5 is the charged capacity in the fifth step, which can be

described as:

$$Q_5 = Q \cdot \left[SOC_5 - \sum_{k=1}^4 \Delta SOC_k \right] \quad (8)$$

where SOC_5 is the SOC at the end of the fifth stage of charging. Based on the SOC-OCV curve, SOC_5 can be expressed by:

$$SOC_5 = f^{-1}(4.2 - I_5 R) \quad (9)$$

where R is the lumped resistance obtained by the parameter estimation model under the current aging state. f represents the functional relationship between OCV and SOC. The total charging time can be obtained through eqs. (7), (8) and (9).

The temperature rise in k-th step can be calculated by eq. (10):

$$\Delta T_k = \frac{\int_{t_{k-1}}^{t_k} I_k^2 R dt + \int_{t_{k-1}}^{t_k} I_k T_c \frac{dU_{ocv}}{dT_c} dt - \int_{t_{k-1}}^{t_k} h S (T_c - T_a) dt}{Cm} \quad (10)$$

where ΔT_k is the temperature rise in k-th step. At the start of charging, the total heat generation is determined by both Joule and entropy change heat generation, and the temperature rise is not obvious, although the current rate is high. The total temperature rise ΔT of the whole charging process is obtained by summing the temperature rise of each stage:

$$\Delta T = \sum_{k=1}^5 \Delta T_k \quad (11)$$

Therefore, the charging optimization problem becomes a performance evaluation problem, because total charging time and temperature rise are both functions of charging current rates. Fitness function and constraints are as eq. (12).

$$\text{Fitness function } f = \alpha \cdot t(I_j) + \beta \cdot \Delta T(I_j)$$

$$\alpha + \beta = 1$$

$$\text{where } I_j = [I_1, I_2, I_3, I_4, I_5]$$

$$\text{Subject to } 0.13A \leq I \leq 2.6A$$

$$2.75V \leq U_c \leq 4.2V$$

$$\Delta T \leq 15^\circ C$$

$$I_j \geq I_k \text{ if } j < k, j, m = 1, 2, 3, 4, 5 \quad (12)$$

where α and β are the weightings of the charging time and temperature rise, respectively; and I_j is the current of stage j. In eq. (12), all constraints related to the optimization problem should be incorporated into the objective function as penalty factors. The output of fitness function is a value between 0 and 1, which is an indicator to evaluate the charging performance with the corresponding charging currents pattern. The value of 0 means that the corresponding charging currents pattern is the least acceptable one, 1 means the most acceptable.

The maximum charging rate of the 18650 battery used in the experiment was 1C (2.6A), and the operating voltage range is between 2.75 V and 4.2 V according to the battery manual. In addition, previous research shows that when the temperature

exceeds 45 °C, the charging process will cause changes in the electrode materials and accelerate the battery aging [21]. In this article, the constraints on charging current, voltage and temperature rise are specified to be in the range [0.13,2.6] A, [2.75,4.2] V and [0,15] °C, respectively.

C. Charging Current Pattern Optimization

Kennedy and Eberhart proposed the PSO algorithm in 1995 [22], and the basic idea of which was derived from the research on the foraging behavior of birds. It is a powerful tool for dealing with global optimization problems with single or multiple objective functions containing several local optimum values. Compared with other optimization algorithms, the most significant feature of the PSO algorithm is that the optimization process is simple and the optimization result is effective, because only two model equations are needed to complete the entire optimization process.

$$v_i(t+1) = \omega v_i(t) + c_1 r_1 (P_{best,i} - x_i(t)) + c_2 r_2 (G_{best} - x_i(t))$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (13)$$

where $x_i(t)$ and $x_i(t+1)$ are the current and the new positions after iteration of the particle, respectively; $v_i(t)$ and $v_i(t+1)$ represent the current and the new velocities after iteration of the particle, respectively; c_1 and c_2 are the cognitive and social learning factors, respectively; r_1 and r_2 are random numbers in the range [0,1], respectively; w is the inertia weight; $P_{best,i}$ is the location at which particle i previously had the best fitness value; and G_{best} is the location with the best global fitness value.

According to the analysis of the electrochemical characteristics of the battery, it is known that the charging time and the charging temperature rise are mainly related to the charging current, and there is a negative correlation between them. In the PSO-based multi-objective optimization problem, the determination of fitness function is of significant importance. In this article, fuzzy reasoning is used to determine the output value of fitness function and evaluate the charging performance. The output also provides a guide for particles updating in the PSO optimization process.

The inputs of the fuzzy membership function are charging time and maximum temperature rise during charging, which are functions of charging current rates. The universe of discourse in the membership functions of inputs is defined on the domains [30 min, 90 min], [3 °C, 15 °C], respectively, to meet the common considerations of the rapid charging criterion. The output represents the acceptability of the charging current scheme, and its domain is [0,1], the value of 0 means that the corresponding charging currents pattern is the least acceptable one, 1 means the most acceptable.

In the charging process, the charging time should be as short as possible, and the maximum temperature rise should be the smallest, which is the most acceptable charging effect. On the contrary, long charging time and large maximum temperature rise are unacceptable effects. In the fuzzy controller, the Mamdani-type minimum inferential method cooperating with the center-of-area (COA) defuzzification procedure is utilized to produce the crisp evaluator output. The schematic of fuzzy control is shown in Fig. 10.

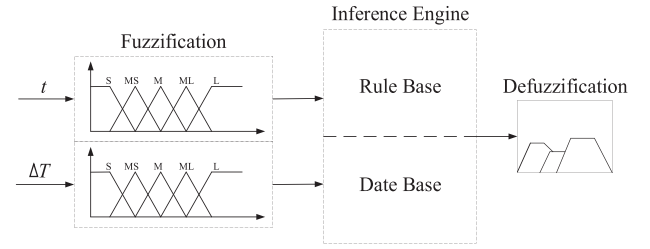


Fig. 10. The schematic of fuzzy inference.

TABLE II
FUZZY RULES OF THE FITNESS FUNCTION

$t \backslash \Delta T$	S	MS	M	ML	L
S	VL Rule1	LL Rule2	L Rule3	ML Rule4	M Rule5
MS	LL Rule6	L Rule7	ML Rule8	M Rule9	MS Rule10
M	L Rule11	ML Rule12	M Rule13	MS Rule14	S Rule15
ML	ML Rule16	M Rule17	MS Rule18	S Rule19	SS Rule20
L	M Rule21	MS Rule22	S Rule23	SS Rule24	VS Rule25

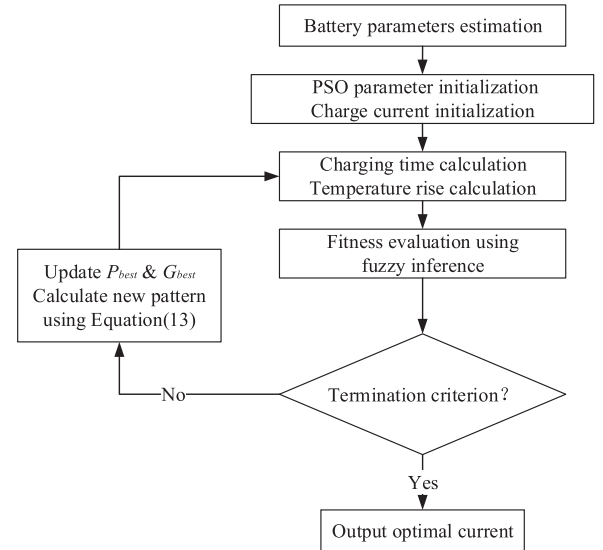


Fig. 11. Flowchart of the optimization process.

As seen in Fig. 10, each input variable t and ΔT is mapped into five different linguistic values, S, MS, M, ML, L. Therefore, the proposed fuzzy rule table will contain 25 different rules, which is shown in Table II.

As shown in Table II, when the charging time is short and the temperature rise is small, corresponding to Rule 1, the fitness function obtains the maximum value (Very Large, VL). When the charging time is long and the temperature rise is large, corresponding to Rule 25, the fitness function obtains the minimum value (Very Small, VS).

As shown in Fig. 11, the main operating procedures are described as follows.

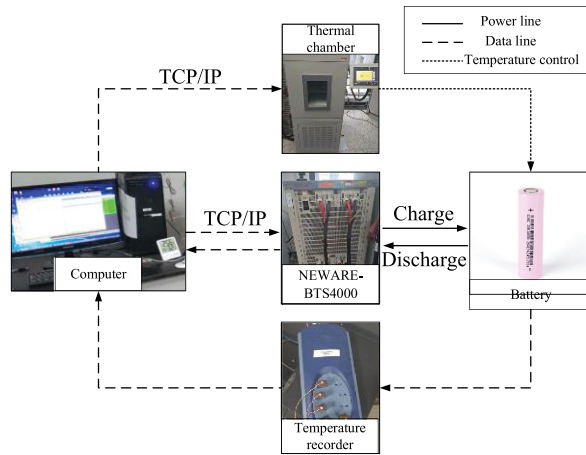


Fig. 12. Experimental platform.

- Step 1:* Extract the aging characteristic parameters, according to the IC curve of the different aging states. The capacity and lumped resistance are estimated using the proposed method.
- Step 2:* Initialize the parameters of the PSO algorithm, including inertial weight and learning factors c_1 and c_2 . The initial charging patterns of each battery for stage 1 are randomly generated.
- Step 3:* Calculate the charging time and charging temperature rise respectively according to eqs. (10) and (13).
- Step 4:* The charging time and charging temperature rise as taken as input of fuzzy reasoning to determine the fitness value of each particle in the swarm.
- Step 5:* Two convergence conditions are set in this article. When both conditions are satisfied, the optimal solution is output. The two convergence conditions are as follows: (1) the standard deviation of the fitness function value of all particles is less than 0.01; (2) the difference between the optimal fitness function value in the current iteration and the optimal fitness function value in the previous iteration is less than 0.0005. When convergence condition (1) is satisfied, the continuous iteration loses its meaning. When condition (2) is satisfied, it means that all particles have reached the extreme value, and the optimization can be concluded.
- Step 6:* If the convergence condition is not reached, the variable is updated according to eq. (13), and Steps 3 to 5 are repeated until the convergence condition is met and the optimization result is output.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

The experimental setup is shown in Fig. 12. The instrumentation used in this article includes a NEWARE-BTS4000 testing system, GDW-50 thermal chamber, TC-08 temperature recorder, NMC 18650 batteries and a PC. Their parameters have been illustrated in Section II.

The optimized charging current pattern is programmed into NEWARE-BTS4000 to charge the battery accordingly, and the battery voltage and current are recorded during charging. The temperature recorder is used to measure and record the surface

TABLE III
BATTERY PARAMETER ESTIMATION RESULTS

Battery number	Capacity (Ah)	Resistance (mΩ)	SOH (%)
1	1.75	112	64.4
2	2.38	86	90.2
3	2.63	73	100

temperature of the battery in real time and to send the recorded temperature data back to the computer. The thermal chamber controls the ambient temperature during the battery charging process to reduce the error of experimental results caused by the variation in ambient temperature.

B. Experimental Results

In order to verify the proposed charging strategy, two experiments were conducted. In the first experiment, the proposed charging strategy was verified with three batteries without knowing the initial state and the aging state. The capacity and lumped resistance obtained using the proposed ICA and PCA based estimation method are shown in Table III. The SOH is the ratio of the current battery capacity to that of a new battery.

The five-stage constant-current charging patterns are optimized using the proposed charging optimization strategy, and the charging patterns for batteries No.1(64.4%SOH), No.2(90.2%SOH) and No.3(100%SOH) are [2.016,1.729,1.320,1.242,1.150],[2.190, 1.865, 1.688,1.431,1.211] and [2.307,2.105,1.706,1.527,1.373], respectively. The comparison between simulation and experiments is shown in Table IV.

As is shown in Table IV, the estimated and experimental results of charging time and maximum temperature rise are almost consistent for batteries with different SOHs. When battery ages, the charging time decreases, because of the loss of capacity and the increase of resistance. Besides, the maximum temperature rise is not obvious for batteries with different SOHs, because more attention is paid to maintaining the temperature rise in the proposed charging strategy. The estimation error of maximum temperature rise is within 1 °C. It proves that the proposed charging optimization strategy can effectively solve the problem of charging for aged batteries, taking charging time and temperature rise into consideration.

To further verify the advantages of the proposed charging strategy, the comparison between the proposed method and the traditional CC-CV charging method is conducted. The temperature variations during charging for three batteries with different SOHs are shown in Fig. 13. The experimental data of each battery for two methods are shown in Table V.

The maximum temperature rise improvement of the proposed method is obvious comparing with CC-CV method. The proposed method can maintain the maximum temperature rise within 5 °C for batteries with unknown initial parameters and SOHs. While the maximum temperature rise of the traditional method increases dramatically, because of the resistance increase cause by aging. The proposed charging strategy takes charging time and temperature rise into consideration to optimize the five-stage charging current pattern.

In order to compare the effect of charging time, an extended charging is conducted with 1/3C to make the battery fully

TABLE IV
CHARGING OPTIMIZATION RESULTS

Battery number	SOH (%)	Charging Time (h)		Max temperature rise (°C)	
		Estimated	Experimental	Estimated	Experimental
1	64.4	0.988	1.044	3.8	4.6
2	90.2	1.221	1.244	3.9	4.2
3	100	1.306	1.366	4.5	4.4

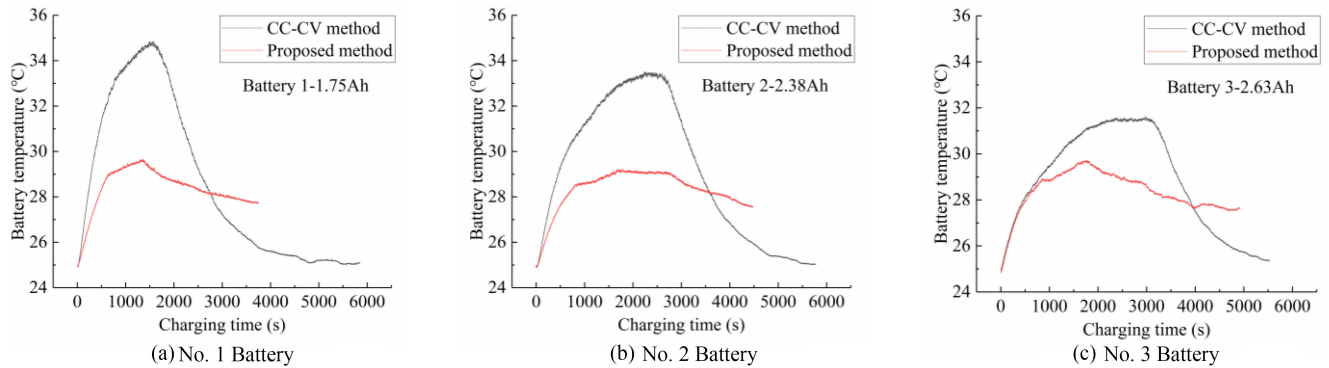


Fig. 13. Battery temperature variation during charging for three batteries.

TABLE V
CHARGING RESULTS FOR TWO METHODS

Battery number	Charging method	Charging time (h)	Max temperature rise (°C)	Charged capacity (Ah)
1	Proposed method	1.044(1.297Ext)	4.6	1.53(1.75 Ext)
	CC-CV method	1.625	9.9	1.75
2	Proposed method	1.244(1.555Ext)	4.2	2.1(2.37 Ext)
	CC-CV method	1.604	8.5	2.37
3	Proposed method	1.366(1.573Ext)	4.4	2.45(2.63 Ext)
	CC-CV method	1.536	6.6	2.63

charged at different SOHs. And the extended charging time is also added to the total charging time with “Ext” marked in Table V. Thus, the charging time of the proposed method can be compared with the CC-CV method. It can be seen that when the battery is new (Battery Number 3), the charging time is slightly larger than CC-CV method, but the reduction effect of the maximum temperature rise of the battery is obvious. Comparing the extended charging time with CC-CV charging time, the reduction of both charging time and maximum temperature can be achieved for aged battery. It can also be seen that the maximum temperature rise and charging time are reduced by 5.3 °C and 0.328h, respectively, for Battery Number 1 with 64.4%SOH, comparing with the CC-CV method.

V. CONCLUSION

Charging performance is of significant importance to a battery, especially for an aged battery. A new charging strategy for aged batteries that combines the PSO optimization algorithm with fuzzy reasoning and takes into consideration the charging time and charging temperature rise is proposed in this article. The

characteristic parameters of different aging states are obtained through HPPC experiments offline. In addition, the peak values and positions on the ICA curve are analyzed to estimate battery aging state parameters, which uses the PCA method to estimate the capacity and lumped resistance of batteries with different SOHs. A battery electric-thermal model is established, and it is proved that the increase in internal resistance caused by aging contributes to heat generation. The experimental results show that the proposed method is able to reduce charging time and temperature rise by optimizing the charging current of five-stage charging. The proposed method is also compared with traditional CC-CV charging method. For three target batteries with unknown SOHs and initial states, the proposed charging strategy can maintain the maximum temperature rise within 5 °C, and improve charging speed. The charging strategy provides a way to effectively charge an aged battery.

In our future work, we will focus on two aspects. Firstly, developing online SOH estimation method to make this charging optimization working in real time. Secondly, more battery samples at different aging states will be utilized to verify the proposed charging optimization method.

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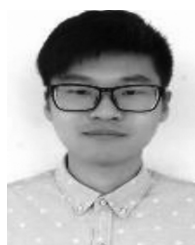
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